

Android Overview

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- Android operating system
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Android operating system

- Android is a Linux-based operating system
- Open source
 - <https://source.android.com/>
 - Just the OS, not the Play store or Google play services
- User Interface for touch screens
- Highly customizable for devices / by vendors

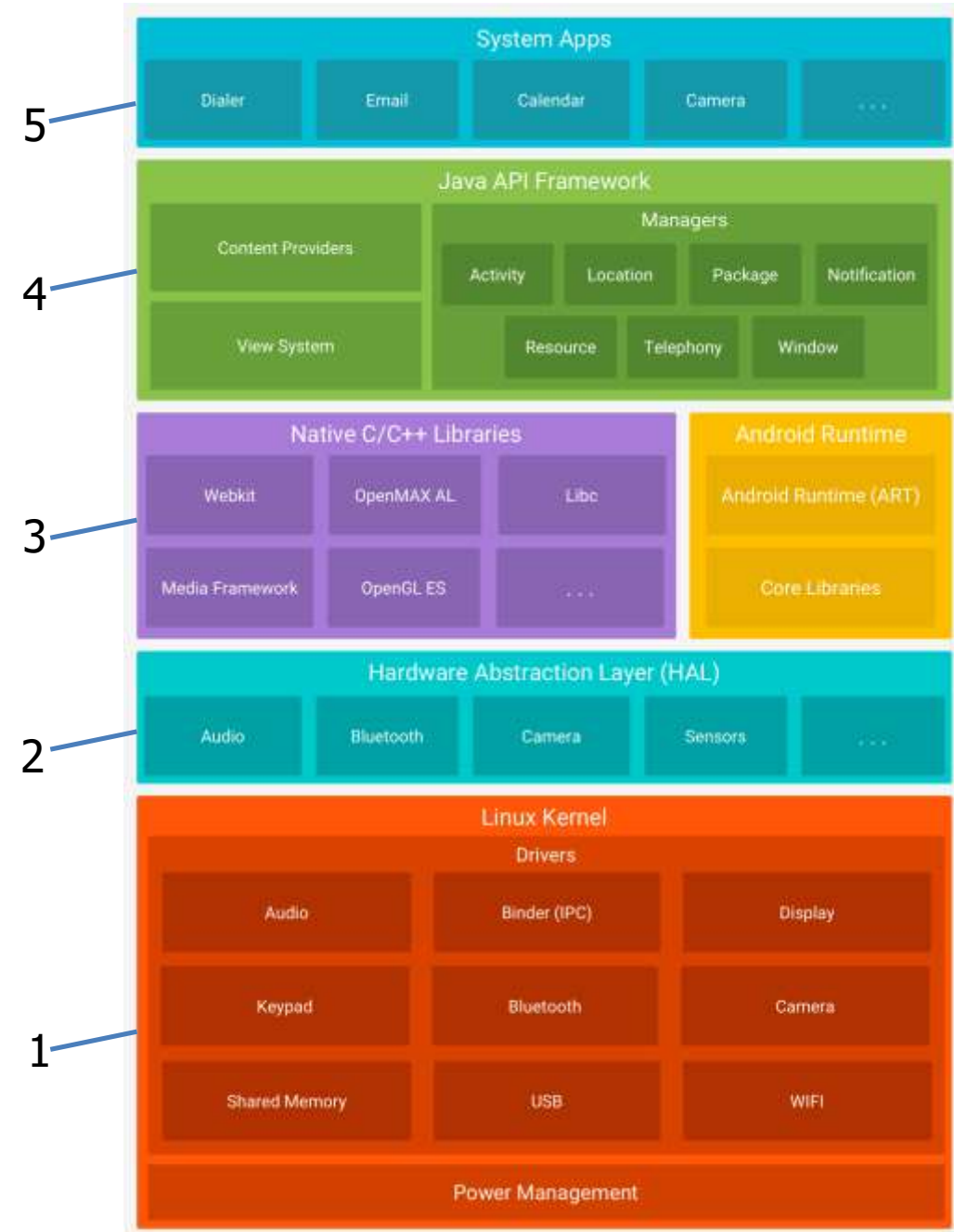


Android Ecosystem

- Android - phones and tablets
- Android wear - smartwatch
 - Extended notification center
- Android TV - televisions, consoles
 - Native TV app experience for IPTV
- Android auto - cars
 - Connecting phone with the car
- Android things
 - Platform for IoT
- Google glass

Android Platform Architecture

1. Linux Kernel
2. HAL: Expose device hardware capabilities
3. Expose native APIs; run apps
4. Java API Framework: Android OS API in Java framework
5. System and user apps



Android Platform Architecture

1. Linux Kernel

- Threading and low-level memory management
- Security features
- Drivers

Android Platform Architecture

2. Hardware Abstraction Layer (HAL): Standard interfaces that expose device hardware capabilities as libraries

Examples: Camera, bluetooth module

3.1 C/C++ libraries: give access to core native Android system components and services.

3.2. Android runtime: Each app runs in its own process with its own instance of the Android Runtime.

Android Platform Architecture

4. Java API Framework:

The entire feature-set of the Android OS is available to you through APIs written in the Java language.

- View class hierarchy to create UI screens
- Notification manager
- Activity manager for life cycles and navigation

Android Platform Architecture

5. System and user apps

- System apps have no special status
- System apps provide key capabilities to app developers
- Example:

Your app can use a system app to deliver a SMS message.

Android versions

Code name	Version numbers	API level	Release date
No codename	1.0	1	September 23, 2008
No codename	1.1	2	February 9, 2009
Cupcake	1.5	3	April 27, 2009
Donut	1.6	4	September 15, 2009
Eclair	2.0 - 2.1	5 - 7	October 26, 2009
Froyo	2.2 - 2.2.3	8	May 20, 2010
Gingerbread	2.3 - 2.3.7	9 - 10	December 6, 2010
Honeycomb	3.0 - 3.2.6	11 - 13	February 22, 2011
Ice Cream Sandwich	4.0 - 4.0.4	14 - 15	October 18, 2011
Jelly Bean	4.1 - 4.3.1	16 - 18	July 9, 2012
KitKat	4.4 - 4.4.4	19 - 20	October 31, 2013
Lollipop	5.0 - 5.1.1	21- 22	November 12, 2014
Marshmallow	6.0 - 6.0.1	23	October 5, 2015
Nougat	7.0	24	August 22, 2016
Nougat	7.1.0 - 7.1.2	25	October 4, 2016
Oreo	8.0	26	August 21, 2017
Oreo	8.1	27	December 5, 2017
Pie	9.0	28	August 6, 2018
Android 10	10.0	29	September 3, 2019
Android 11	11	30	September 8, 2020

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Android 12	12	31	October 4, 2021
Android 12L	12.1	32	March 7, 2022
Android 13	13	33	August 15, 2022
Android 14	14	34	October 4, 2023

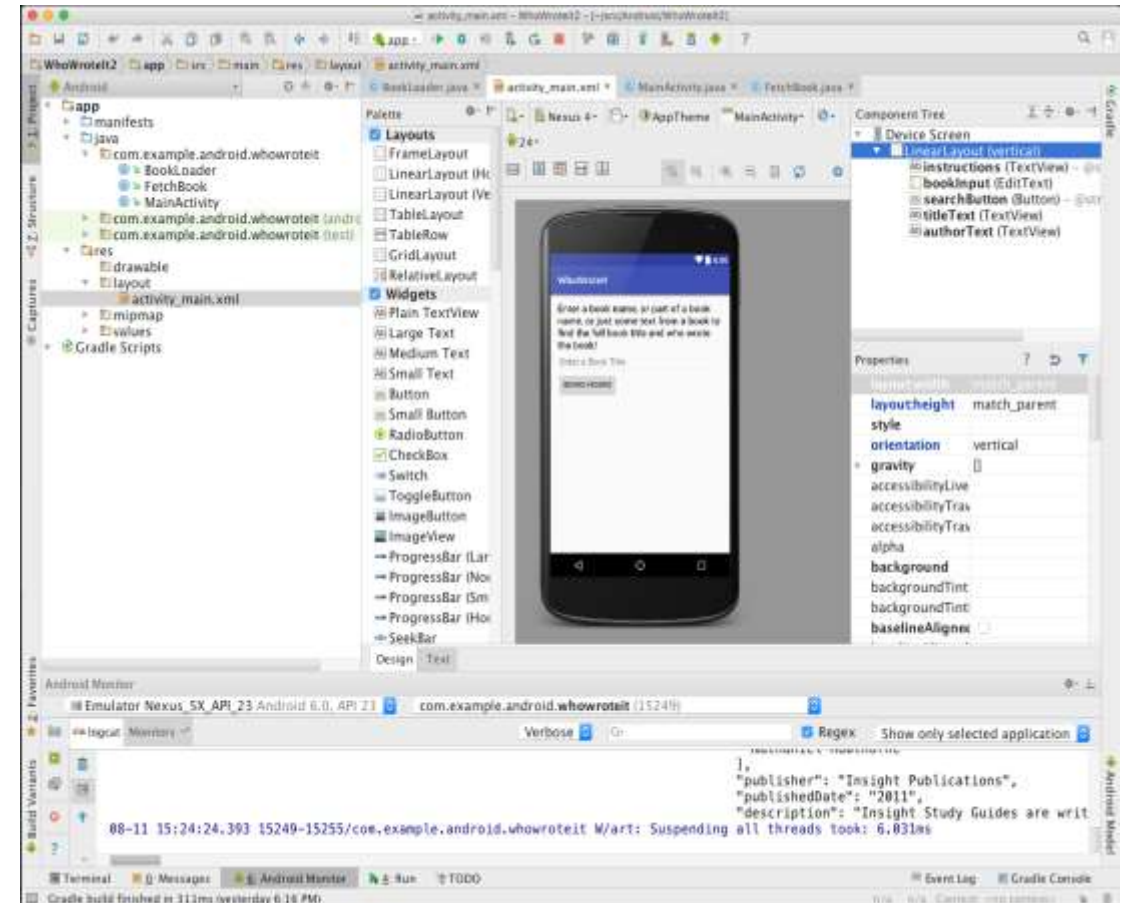
Challenges of Android development

- Multiple screen sizes and resolutions
- Performance: make your apps responsive and smooth
- Security: keep source code and user data safe
- Compatibility: run well on older platform versions

Android application development

• Android Studio

- Develop, run, debug, test, and package apps
- Monitors and performance tools
- Virtual devices
- Project views
- Visual layout editor



Android application development

- Android Software Developer Kit (SDK)
 - Development tools (debugger, monitors, editors)
 - Libraries (maps, wearables)
 - Virtual devices (emulators)
 - Documentation (developers.android.com)
 - Sample code

References

- <https://google-developer-training.github.io/android-developer-fundamentals-course-concepts-v2/>

Practice Project samples for groups (5 members)

- **Describe:** Develop an Android application for Products Sale
- **Database:** Using MySQL/SQLServer (using Restful API) or database on Firebase cloud
- **Functions:**
 - Sign Up/Login screen: manage users
 - List of Products screen: show list of products
 - Product details screen: show information of a product
 - Product Cart screen: show products that user selects to buy
 - Billing screen: process payment of user
 - Show a notification if Cart has products when opening the application
 - Map screen: show Store's location
 - Chat screen: customers can chat with Store

Other apps

- Task management
- Expense management
- Story reading app
- Music app
- Video app
- Weather app
- Some games
- Etc.

Requirement for project

- Each member at least 3-4 screen (medium)
- Submit source code & Document
- Present the result.